

UNIT CODE	H7DT
UNIT NAME	WEB PROGRAMMING I
ASSESSMENT TYPE	WEB APPLICATION DEVELOPMENT PROJECT, RECORDED SOLUTION DEMONSTRATION, AND LIVE VIVA VOCE EXAMINATION WITH SCENARIO-BASED CODING CHALLENGE

ASSESSMENT GUIDE

OVERVIEW

This assessment evaluates the student's ability to analyse, design, develop, implement, explain and defend a complete web application solution based on a given business scenario.

Students are required to:

- Develop a complete web application solution.
- Submit a recorded solution demonstration.
- Attend a Live Viva Voce Examination, Project Defence and Scenario-Based Coding Challenge.

The assessment is designed to evaluate both theoretical understanding and practical web application development skills while verifying the authenticity and ownership of submitted work.

OVERALL ASSESSMENT WEIGHTING

COMPONENT	MARKS
Component A – Web Application Development Project	30
Component B – Recorded Solution Demonstration	20
Component C – Live Viva Voce Examination, Project Defence and Scenario-Based Coding Challenge	50
TOTAL	**100 MARKS**

IMPORTANT NOTE

Assessment marks will be awarded based on the student's ability to:

- Develop an appropriate web application solution.
- Apply web development concepts, methodologies, and techniques.
- Analyse requirements and design decisions.
- Develop static and dynamic web pages.
- Implement PHP functionality correctly.
- Implement database connectivity and data manipulation.
- Use multimedia and file uploading mechanisms.
- Explain, justify, and defend submitted work.
- Demonstrate ownership and authenticity throughout the assessment process.

SCENARIO

CampusHub is a growing student services organisation that coordinates student clubs, events, workshops, competitions, sports activities, and student communities across multiple educational institutions.

The organisation currently manages event registrations, member records, activity schedules, event photographs, announcements, and attendance information using spreadsheets, emails, and social media platforms.

Management has decided to introduce a web-based information system that allows students to:

- Register for events online.
- View upcoming events and activities.
- Manage member profiles.
- Upload photographs and activity updates.
- Access multimedia content.
- Submit online forms.
- View announcements and notifications.

Administrators must be able to:

- Manage student records.
- Manage events and activities.
- Manage registrations.
- Upload media content.
- Publish website content.
- Maintain organisational information.

Students are required to analyse the requirements and develop a suitable web application solution for the organisation.

Reasonable assumptions may be made where necessary. All assumptions must be clearly documented.

COMPONENT A – WEB APPLICATION DEVELOPMENT PROJECT

OVERVIEW

Students are required to develop a complete web application solution for the given scenario.

DEVELOPMENT REQUIREMENTS

TASK	REQUIREMENTS
Requirements Analysis	Identify users, features, business processes, assumptions, and system requirements.
Website Planning	Define purpose, scope, navigation structure, and expected benefits.
User Interface Design	Design page layouts, menus, navigation structure, header, footer, and content presentation.
HTML Development	Develop static and dynamic web pages using appropriate HTML elements.
CSS Implementation	Apply external CSS, selectors, formatting, and reusable styling techniques.
PHP Development	Implement variables, functions, forms, parameter passing, includes, and dynamic content.
Database Connectivity	Connect the application to a database using PHP.
Data Manipulation	Implement Insert, Update, Delete, and Select operations.
XML Implementation	Demonstrate the use of XML technologies where appropriate.
Multimedia Integration	Implement audio and/or video content.
File Upload Functionality	Implement file uploading features.
Error Handling	Implement validation and error handling mechanisms.

WEB APPLICATION IMPLEMENTATION SCOPE

CATEGORY	REQUIREMENTS
HTML	Static and Dynamic Web Pages
CSS	External CSS, Styling, Layout Design, Selectors
PHP	Variables, Functions, Forms, Parameter Passing, Includes

Database Connectivity	Database Connection and CRUD Operations
XML	XML Syntax and Validation
Multimedia	Audio and Video Integration
File Uploading	Upload and Display Files
Error Handling	Validation and Error Management

SUBMISSION FORMAT

SECTION	CONTENT REQUIREMENTS
Cover Page	Project Title, Student Name, Student ID, Batch, Unit Name, Submission Date
1. Introduction	Introduction to the scenario and system purpose
2. Requirements Analysis	Users, business requirements, assumptions, and objectives
3. Website Planning	Scope, navigation structure, and design decisions
4. User Interface Design	Page layouts, menus, navigation, header, and footer
5. HTML and CSS Development	Implementation approach and screenshots
6. PHP Development	Variables, functions, forms, parameter passing, and imports
7. Database Connectivity and Data Manipulation	Database design and CRUD implementation
8. XML Implementation	XML usage and implementation
9. Multimedia and File Upload Features	Feature implementation and screenshots
10. Error Handling and Validation	Validation and error handling mechanisms
11. Conclusion	Summary of the solution
Appendices	Source Code Screenshots, Database Screenshots, Supporting Evidence

SUBMISSION REQUIREMENTS

ITEM	FORMAT
Web Application Report	PDF
Source Code	ZIP
Database Script	.sql
Supporting Screenshots	Included within Report
Recommended Report Length	10–15 Pages (excluding appendices)

ASSESSMENT EVALUATION MATRIX

ASSESSMENT AREA	MARKS
Requirements Analysis and Website Planning	05
User Interface Design and Navigation Structure	05
HTML, CSS and PHP Implementation	10
Database Connectivity and Data Manipulation	05
XML Implementation	05
Multimedia, File Uploading and Error Handling	05
TOTAL	**30 MARKS**

COMPONENT B – RECORDED SOLUTION DEMONSTRATION

OVERVIEW

Students are required to prepare and submit an individual self-recorded video presentation explaining the web application solution developed for the assessment scenario.

The purpose of the recording is to demonstrate the student's understanding of the submitted solution and provide evidence of ownership prior to the Viva Voce Examination.

PRESENTATION REQUIREMENTS

Students must:

- Display their National Identity Card (NIC), Passport, Driving Licence, or other approved photographic identification at the beginning of the recording.
- Appear clearly on camera throughout the recording.
- Use their own voice throughout the presentation.
- Demonstrate personal understanding of the submitted solution.
- Clearly explain the work developed.

Medium: English

PRESENTATION CONTENT

Students should explain:

- Requirements Analysis
- Website Structure and Navigation
- HTML and CSS Development
- PHP Functionality
- Database Connectivity
- XML Implementation
- Multimedia Features
- File Upload Functionality
- Error Handling
- Assumptions Made
- Key Design Decisions

Students may use screen sharing, diagrams, source code, database structures, or other supporting materials where appropriate.

SUBMISSION FORMAT

- Video Format: MP4
- Duration: 5–10 Minutes

ASSESSMENT EVALUATION MATRIX

ASSESSMENT AREA	MARKS
Explanation of Requirements and Design	05
Explanation of Web Application Functionality	05
Explanation of Database and Multimedia Feature	05
Communication, Professional Presentation, Authenticity and Ownership	05
TOTAL	20 MARKS

COMPONENT C – LIVE VIVA VOCE EXAMINATION, PROJECT DEFENCE AND SCENARIO-BASED CODING CHALLENGE

OVERVIEW

Students are required to attend a Live Viva Voce Examination, Project Defence, and Scenario-Based Coding Challenge.

The purpose of the assessment is to verify:

- Knowledge and understanding.
- Practical web development skills.
- Application of web programming concepts.
- Problem-solving ability.
- Ownership and authenticity of submitted work.
- Ability to respond to new technical requirements.

DURATION

- Approximately 20–30 Minutes

VIVA VOCE REQUIREMENTS

Following the submission of Components A and B, students may request to undertake the Viva Voce Examination through the Student Portal before the published application deadline.

Students must be prepared to:

- Explain the submitted web application.
- Explain HTML implementation.
- Explain CSS implementation.
- Explain PHP functionality.
- Explain database connectivity and CRUD operations.
- Explain XML implementation.
- Explain multimedia integration.
- Explain file uploading mechanisms.
- Discuss assumptions made.
- Justify technical decisions.
- Defend design choices.

SCENARIO-BASED CODING CHALLENGE

Students may be presented with one or more practical scenarios during the Viva Voce Examination.

Examples may include:

- Add a new registration form.
- Create a new PHP function.
- Modify an existing web page layout.
- Implement a search feature.
- Add a file upload mechanism.
- Add a multimedia feature.
- Implement form validation.
- Handle an error condition.
- Create a database connection.
- Develop a database query.
- Improve website navigation.

Students may be required to:

Explain implementation approaches.

- Write code snippets.
- Analyse requirements.
- Debug existing code.
- Identify errors.
- Justify solutions.
- Recommend alternative approaches.

ASSESSMENT EVALUATION MATRIX

ASSESSMENT AREA	MARKS
Knowledge and Understanding of Web Programming Concepts	05
Application of HTML, CSS, PHP and XML Technologies	05
Database Connectivity and Data Manipulation	05
Analysis of Requirements and Design Decisions	05
Scenario-Based Coding Challenge and Problem Solving	15
Evaluation, Justification and Defence of Technical Decisions	10
Ownership and Authenticity of Submitted Work	05
TOTAL	50 MARKS

APPLICATION FOR THE VIVA VOCE EXAMINATION

Students are responsible for applying for the Viva Voce Examination through the Student Portal before the published application deadline.

Failure to apply within the published deadline will result in the student being recorded as Absent and the assessment attempt being deemed unsuccessful.

The student will be required to undertake a repeat assessment, which may be subject to repeat assessment fees and a maximum achievable grade of Pass.

Students are therefore strongly advised to apply before the published application deadline.

ACADEMIC INTEGRITY AND AUTHENTICITY

Students are expected to maintain the highest standards of academic integrity throughout the assessment process.

All submitted documents, source code, database scripts, recorded presentations, and Viva Voce responses must represent the student's own understanding and work.

Students may be required to participate in additional verification activities, demonstrations, or oral questioning to verify authenticity, ownership, and academic integrity.

Failure to demonstrate reasonable ownership and understanding of submitted work may result in academic review procedures in accordance with institutional regulations.